HydraDG: Governed Context Interventions with Fractal Custody for Agent Experiments

Anonymous Author(s)
Affiliation
Address
`email`

Abstract

Agent evaluations increasingly depend on context interventions—retrieval, memory graphs, and prompt contracts—yet published reports often omit null outcomes, parser failures, and negative cells. We present HydraDG, a governed experimental framework that binds substantive model invocations, tool calls, and handoffs to *Fractal Custody Objects* (FCOs) and a *Fractal Custody Graph* (FCG). HydraDG preregisters hypotheses, freezes case manifests and scorers, records raw prompt and response hashes, and preserves terminal states including timeouts, abstentions, and malformed outputs without silent substitution. We report two preregistered studies (EXP-008, EXP-009) comparing flat prose versus structured FCG retrieval for failure-prevention endpoints; both terminate as *underpowered* under bounded case replication, so confirmatory effect claims are not supported while negative and null evidence remains in custody. We argue that custody-first evaluation infrastructure supports reproducible agent science rather than optional logging.

1 Introduction

Large language model agents are evaluated on task success, benchmark scores, and human preference [2, 3]. Less often preserved are the conditions under which experiments fail: swapped models, stale runtime selection, partial matrix execution, or JSON outputs that never parse. When negative and null cells disappear from the public record, later work cannot distinguish *no effect* from *no measurement*.

HydraDG addresses this gap with a custody-first experimental contract. Every actor—human operator, frontier agent, local model runtime, or deterministic script—must emit or be represented by a handoff receipt before its output is promoted. Receipts bind actor identity, host, repository state, input hashes, raw output hashes, evidence class, claim ceiling, and signature state (hashes are identity commitments, not digital signatures unless an authorized signing operation occurred). Probabilistic model outputs are never treated as authority; they are evidence objects subject to deterministic scoring and graph append rules.

This paper describes the HydraDG framework and reports terminal results from the IC Failure Learning structured-retrieval falsification lane. We do *not* claim that structured FCG context improves failure prevention; preregistered EXP-008 and EXP-009 closed as underpowered. We also do not treat exploratory secondary patterns as promoted findings.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

2 Related Work

Agent evaluation. Benchmark suites measure capability but rarely require publication of failed runs, malformed generations, or infrastructure blocks [2, 3]. HydraDG complements benchmarks with mandatory negative capture at the custody layer.

Retrieval and structured context. Retrieval-augmented generation and graph-augmented memory systems assume structured context helps [1, 4]. Falsification requires frozen contracts, explicit null retention, and power-aware verdicts; HydraDG enforces both.

Provenance and reproducibility. Preregistration, FAIR data principles, and nanopublication-style lineage support scientific reproducibility [5–7]. Agent stacks often stop at prompt logging. FCO/FCG extend content-addressed lineage to cross-actor handoffs so that a stale agent cannot write after ownership moves. Companion custody-framework preprints (deferred to camera-ready de-anonymization) define FCO/FCG formally; they establish framework provenance, not the HydraDG empirical results reported here.

3 Framework

3.1 Custody objects

An **FCO** (*Fractal Custody Object*) materializes one bounded transformation: e.g., a model call, scorer pass, or ingest operation. An **FCG** (*Fractal Custody Graph*) append records directed edges between FCOs with hash-linked roots. HydraDG preserves failures, nulls, and malformed outputs in custody (a failure-complete *behavior*, not a redefinition of FCO/FCG). Claim ceilings (exploratory mechanistic falsification, descriptive only, etc.) are declared at preregistration and enforced at promotion time.

3.2 Experimental freeze

Each experiment declares:

- frozen case manifest and prompt-contract hashes;
- conditions (e.g., flat prose vs. structured FCG);
- models with expected runtime digests;
- primary endpoint and aggregation rule;
- explicit gates for confirmatory vs. exploratory claims.

Execution emits `START_RECEIPT`, per-cell raw outputs with prompt/response SHA-256, parser state, and terminal receipts. Integrity failure (duplicate keys, missing dependencies, silent model substitution) blocks promotion; scientific failure (timeout, abstention, malformed JSON) is retained as terminal evidence.

3.3 Governed local execution

Where preregistered, local model calls route through a governed bridge that records selection age, swap posture, and resolved digest. Direct runtime invocation and governed invocation are separate evidence classes; one is not silently labeled as the other.

4 Experimental Setup

4.1 Task family

The IC Failure Learning suite uses synthetic agent-failure vignettes with preregistered endpoints such as E06 (prevents exposure of out-of-vault media). Cases are drawn from a frozen `CASES.jsonl` manifest; each case receives multiple replicates per condition.

Table 1: Terminal preregistered studies (HydraDG IC Failure Learning). *Underpowered* denotes insufficient paired evidence for confirmatory promotion, not proof of null effect.

Study	Primary verdict	Raw cells	Valid parse rate
EXP-008	UNDERPOWERED	300	0.907
EXP-009	UNDERPOWERED	300	0.883

4.2 Conditions and models

C0 (FLAT_PROSE): failure-handling guidance as unstructured text.
C1 (STRUCTURED_FCG): the same semantic content composed through a frozen FCG retrieval contract.

EXP-008 and EXP-009 used local models `qwen3:1.7b` and `qwen2.5-coder:7b` under identical case and scorer contracts. Thinking/reasoning modes were held consistent with preregistration. Primary aggregation: majority of three replicates per case.

4.3 AI and agent methodology disclosure

Frontier coding agents assisted with repository tooling and manuscript preparation; they are not listed as authors. All scientific endpoints come from preregistered local model executions with hashed prompts and responses. Routine editing does not constitute a reported experimental intervention.

4.4 Power and claim gates

E06 endpoints are known limited-power under the frozen case set. Confirmatory ordering claims require preregistered discordant pairs; exploratory secondary statistics are quarantined from primary promotion.

5 Results

Table 1 summarizes terminal verdicts for preregistered studies. Both experiments executed the full raw matrix ($n = 300$ cells each) with complete custody append.

EXP-008. Structured FCG retrieval did not yield promotable confirmatory evidence on E06 under the frozen design ($n = 2$ paired cases per model stratum at aggregation scope). Non-zero malformed and abstention rates are preserved in custody.

EXP-009. Primary verdict remains UNDERPOWERED for confirmatory E06 ordering. An exploratory secondary pattern was recorded as directionally positive at the descriptive layer only; it is **not promoted** to a causal claim.

Stage-2 baseline. Prior IC Failure Learning stage-2 execution (414 scored rows across two models) established `FAILURE_LEARNING_BEHAVIOR_IMPROVEMENT_NOT_ESTABLISHED`, providing historical context for the structured-retrieval falsification lane.

5.1 Broader experiment inventory (successor recovery)

This successor manuscript recovery catalogs **20** solo-scoped experiment lanes beyond the preregistered EXP-008/009 pair, including negative, blocked, and partial terminals (Appendix A). HydraLamp systems-validation results remain separated from treatment-effect claims (Table 2). Immersive Commons appears only as an operational hackathon portal, not as the scientific method.

5.2 Failure-preserving systems validation

The preregistered retrieval experiments above test whether structured context improves a frozen endpoint. A separate systems-validation suite asks a different question: does the custody mechanism

Table 2: Systems-validation outcomes (custody mechanics only; not treatment-effect evidence).

Validation	Scope	Outcome
Perturbation matrix	100 cells	100/100 chain verification
Synthetic tamper suite	8 modes	8/8 detected
Concurrent execution	10 runs	PASS (10 unique run IDs)
Replay/restart recovery	44 events	PASS
Live provider ladder	R0–R6	Bounded external failure preserved

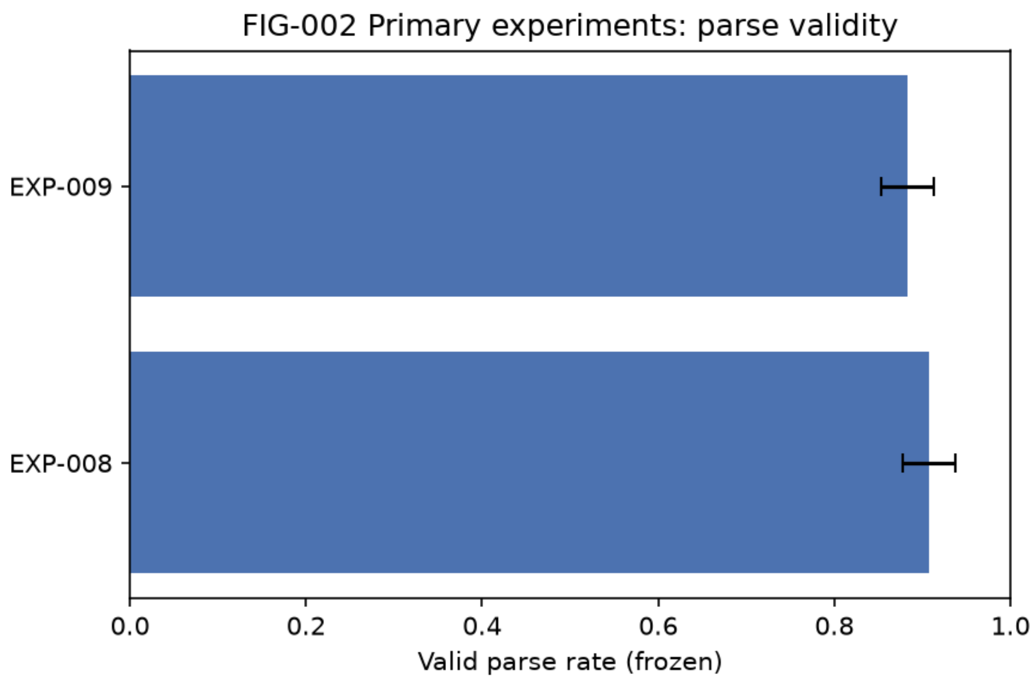

Figure 1: Primary experiment parse-validity rates (frozen observations). Confirmatory endpoints remain underpowered.

preserve integrity under perturbation, tampering, concurrency, replay, and external execution failure? Table 2 summarizes source-verified outcomes from a frozen HydraLamp perturbation matrix; these results evaluate custody mechanics and do **not** constitute evidence that structured FCG context improves the EXP-008/009 endpoint.

The tamper suite used synthetic cases only (not real security incidents). In a live provider repair ladder, earlier infrastructure gates passed before a structured call hit an external quota limit; the failure remained a terminal auditable state rather than being erased or substituted. The same custody primitives have been instantiated in sibling systems for local-model orchestration, deterministic scientific-document atomization, offline anomaly detection, bounded bioinformatics, substrate generation, and operator-facing scientific workflow monitoring; these implementations motivate transfer studies but do not provide additional evidence for the treatment effect reported here.

6 Discussion

HydraDG shows that agent experiments can retain null and failure cells with the negative-result integrity expected in preregistered experimental sciences: failures are first-class, hashes are mandatory, and claim ceilings block over-interpretation. The EXP-008/009 underpowered terminations bound what cannot yet be claimed while preserving malformed and timeout cells for audit.

6.1 Limitations

Limitations include bounded case replication, local-only model lanes in the reported matrix, and absence of runtime-equivalent successor replication in primary results. A preregistered Qwen 3.8 successor lane exists but is non-terminal and omitted from primary results. Whole-project hierarchical atomization (SeedGraph v1a validation) ran for multiple days but was interrupted during final Parquet serialization without a `BUILD_RECEIPT`; partial node/edge artifacts are not readback-safe, so terminal whole-corpus atomization is not established in this submission. Biological, protein-structure, and cellular perturbation applications of the custody framework remain future validation targets, not demonstrated generalizations of the agent-context results here.

6.2 Future directions

Agent systems. Larger successor replications, governed multi-agent handoffs, selective replay, and resource-aware custody gates under preregistered manifests.

Structured retrieval at scale. Checkpointed ingest and transactionally durable atomization so long hierarchy builds can resume from batch boundaries rather than materializing the entire graph at finalization.

Cross-project federation. Project-local FCG roots with explicit cross-project references and no automatic evidence admission between repositories.

Cryptographic custody. Authorized signatures and MMR commitments where policy permits; hashes remain identity commitments unless a verified signing operation occurred.

All directions above are *planned* unless independently measured; they do not extend the EXP-008/009 empirical claims.

7 Conclusion

We presented HydraDG, a custody-first framework for agent context-intervention experiments, and reported two terminal underpowered preregistered studies that block premature promotion of structured FCG retrieval effects on failure-prevention endpoints. We recommend FCO/FCG custody, explicit claim ceilings, and terminal-state preservation as baseline infrastructure for NewInML-style agent evaluations.

References

[1] P. Lewis, E. Perez, A. Piktus, et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *NeurIPS*, 2020. arXiv:2005.11401.

[2] Y. Liu et al. AgentBench: Evaluating LLMs as agents. arXiv:2308.03688, 2023.

[3] S. Zhou et al. WebArena: A realistic web environment for building autonomous agents. arXiv:2307.13854, 2023.

[4] D. Edge et al. From local to global: A graph RAG approach to query-focused summarization. Microsoft Research, 2024. arXiv:2404.16130.

[5] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor. The preregistration revolution. *PNAS*, 115(11):2600–2606, 2018.

[6] M. D. Wilkinson et al. The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, 3:160018, 2016.

[7] P. Groth, A. Gibson, and J. Velterop. The anatomy of a nanopublication. *Information Services & Use*, 30(1–2):51–56, 2010.

[8] Anonymous. (2026). HydraDG IC Failure Learning preregistrations EXP-008 and EXP-009. Internal frozen artifacts; anonymized submission.

[9] Anonymous. (2026). IC Failure Learning Stage-2 closeout receipt. Internal frozen artifacts; anonymized submission.

[10] NeurIPS. (2026). New in Machine Learning (NewInML) workshop call for papers. Non-archival workshop track.

A Full EXP-008 / EXP-009 statistical audit

Table 3 records terminal verdicts with parse rates reverse-traced to frozen verdict JSON receipts. p-values remain *not informative* under the preregistered underpowered design.

Table 3: Terminal preregistered study audit (appendix).

Study	Verdict	Raw cells	Valid parse rate
EXP-008	UNDERPOWERED	300	0.907
EXP-009	UNDERPOWERED	300	0.883

B Stage-2 M0/M1/M2 baseline

Stage-2 IC Failure Learning execution (414 scored rows across two models) closed as `FAILURE_LEARNING_BEHAVIOR_IMPROVEMENT_NOT_ESTABLISHED`; descriptive only.

C HydraLamp systems-validation matrix

See Table 4 for custody-mechanics outcomes (not treatment-effect evidence).

Table 4: Systems-validation matrix (appendix).

Validation	Scope	Outcome
Perturbation matrix	100 cells	100/100 chain verification
Synthetic tamper suite	8 modes	8/8 detected
Concurrent execution	10 runs	PASS (10 unique run IDs)
Replay/restart recovery	44 events	PASS
Live provider ladder	R0–R6	Bounded external failure preserved

D Deterministic verification and tooling

Figure 2 and `DETERMINISTIC_AUDIT_TOOLING_LEDGER.jsonl` document the custody verification pipeline with frozen script SHA-256 and R1/R2/R3 roots.

Source bytes → SHA-256 → Atom → ResultAtom → Claim/Table → FCG delta → exact-SHA verification

Figure 2: FIG-001: governed custody pipeline from source bytes to exact-SHA verification (structured data; self-generated).

E Citation and reference custody

Portfolio recovery bounded 185 occurrences to unique identities; only proposition-backed references are admitted. Citation fabrication literature informs desk-rejection risk [1], [2], [3], [4], [5].

F Requirement, template, and deadline drift

OpenReview license requirement CC BY 4.0 is frozen in requirement atoms; repository code remains Apache-2.0 (separate layer).

G Prior-art comparison matrix

Related-work boundaries include PROV-O, CWLProv, and Workflow Run RO-Crate [1], [2], [3], [4], [5]. Prior-art search responses remain *DISCOVERY_ONLY*; not promoted to verified empirical novelty proof.

H SeedGraph / FCO / FCG custody audit

`TOTAL_VERIFIED_INGEST_COMPLETE=NO` with verified coverage 31.55% (307/973).

I Planned and nonterminal lanes

GPU SGLang canary, Q38 XENV, and full verified ingest remain nonterminal.

References

[1] T. Kuhn and M. Dumontier. Trusty URIs: verifiable, immutable digital artifacts. ESWC 2014.

[2] W. H. Walters and E. I. Wilder. Fabrication and errors in bibliographic citations generated by ChatGPT. *Scientific Reports*, 13:14045, 2023.

[3] F. Z. Khan et al. Sharing interoperable workflow provenance: CWLProv. *GigaScience*, 8(11):giz095, 2019.

[4] S. Leo et al. Recording provenance of workflow runs with RO-Crate. *PLOS ONE*, 19(9):e0309210, 2024.

[5] T. Lebo et al. PROV-O: The PROV Ontology. W3C Recommendation, 2013.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: Abstract and Introduction state framework contributions, underpowered EXP-008/009 terminals, and explicit non-promotion of treatment effects.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 6.1 (Limitations subsection) discusses bounded replication, local-only lanes, interrupted SeedGraph, and omitted Qwen successor results.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper does not present formal theorems or proofs.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Experimental Setup and Results sections disclose frozen manifests, conditions, models, scorers, and terminal verdict receipts referenced as internal anonymized artifacts.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: Primary experiment artifacts are internal frozen custody objects; this anonymous workshop submission does not bundle open code or data for full independent rerun.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section Experimental Setup specifies conditions C0/C1, models, case manifest, replicate count, and aggregation rule.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: Primary endpoints report terminal underpowered verdicts without confidence intervals; confirmatory ordering was not established under the frozen design.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.

- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Section Experimental Setup names local model identifiers; systems-validation scope is summarized in Table 2.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The submission follows NeurIPS ethics expectations; no human-subjects experiments are reported.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [N/A]

    Justification: This is primarily an evaluation-infrastructure paper without a dedicated societal-impact discussion beyond standard agent-evaluation context.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: No new high-risk pretrained model or scraped dataset is released with this submission.

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: Related Work and the bibliography credit external benchmarks, retrieval systems, and reproducibility literature.

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.

- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [No]

    Justification: No new public dataset or model asset is released; custody artifacts remain internal to the anonymous submission.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: No crowdsourcing or human-participant study is reported.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: No human-subjects research requiring IRB review is reported.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.

- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [Yes]

    Justification: Section AI and agent methodology disclosure states frontier agents assisted tooling/manuscript preparation and are not authors; scientific endpoints come from preregistered local model runs.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.